

Bibek Poudel

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Professional Summary

Robotics Engineer with hands-on experience developing autonomous manipulation, robot perception, motion planning, and agentic AI systems for industrial and research robotic platforms. Experienced in designing end-to-end robotic software pipelines spanning computer vision, grasp planning, reinforcement learning, sim-to-real deployment, and AI agent workflows using ROS2, MoveIt 2, MuJoCo, Isaac Sim, Python, C++, and modern LLM frameworks. Demonstrated research experience through robotic system development, cross-platform experimentation, and publication in robotic manipulation.

Education

New York University Abu Dhabi

B.Sc. in Computer Engineering, GPA: 3.91/4.00
Minor in Robotics

Abu Dhabi, UAE
Aug 2022 – May 2026

Relevant Coursework:

- *Robotic Manipulation and Locomotion, Robot Vision, Computer Vision, Simulation Tools for Robotics, Introduction to Haptics and Telerobotics in Medicine, Introduction to Machine Learning, Computer-Aided Design*

Experience

iCAS Lab, NYU Abu Dhabi

Student Research Assistant

Abu Dhabi, UAE
Sep 2025 – Present

- Integrated five state-of-the-art robotic grasp planning algorithms (**GGCNN**, **GR-ConvNet**, **Contact-GraspNet**, **AnyGrasp**, and **GraspGen**) into an end-to-end perception pipeline for the **Unitree Z1 Pro**, enabling systematic evaluation of learning-based grasp generation methods.
- Designed **MVB-Grasp**, a minimum-volume-box filtering framework that improves executable diffusion-based grasp selection, and engineered a complete manipulation pipeline integrating camera calibration, coordinate-frame transformations, grasp planning, inverse kinematics, and real-world grasp execution.
- Benchmarked robotic manipulation performance across five robotic platforms (**Unitree Z1 Pro**, **FR3**, **KUKA iiwa**, **Aubo-i5**, and **AgileX Piper**), evaluating cross-platform robustness and deployment of autonomous grasping pipelines.

Robot Control and Learning Lab, NYU Abu Dhabi

Student Research Assistant

Abu Dhabi, UAE
Apr 2025 – Sep 2025

- Implemented **Policy Improvement with Path Integrals (PI²)** to train dynamic robotic manipulation behaviors, enabling autonomous ball-throwing tasks on the **Franka Research 3** manipulator.
- Developed a high-fidelity **MuJoCo** simulation environment for the **Franka Research 3** and benchmarked variable impedance, admittance, and reinforcement learning-based controllers to evaluate manipulation performance prior to hardware deployment.
- Designed and validated **sim-to-real transfer** pipelines that enabled deployment of learned manipulation policies from simulation to the physical robot, improving the efficiency and reliability of real-world experimentation.

Flexible AI-enabled Mechatronic Systems Lab (FAMS)

Research Assistant

New York, USA

Jan 2025 – May 2025

- Designed, fabricated, and assembled a **3-DOF robotic arm** with custom **soft robotic fingers** using **SolidWorks**, silicone molding, and embedded flex sensors, enabling compliant object manipulation and tactile sensing.
- Engineered a wireless real-time robotic control system using **Raspberry Pi**, **ROS**, and **Python**, integrating sensing, communication, and actuation for autonomous manipulation experiments.

Publications

- **Poudel, B.**, Basit, A., and Shafique, M., “MVB-Grasp: Minimum-Volume-Box Filtering of Diffusion-based Grasps for Frontal Manipulation,” *International Joint Conference on Neural Networks (IJCNN)*, 2026.

Selected Robotics Projects

Optimal Grasping and Manipulation using Unitree Z1 Pro Robotic Arm: MuJoCo | Robot Vision | Unitree Z1 | Hand-Eye Coordination | Camera Calibration | Kinematics  [Link](#)

RL-Based Learning for Throwing a Ball with a 7-DOF Robotic Arm: MuJoCo | Reinforcement Learning | Franka Panda | Dynamics | Robot Control | Kinematics  [Link](#)

Pick and Place using Franka Research 3: MuJoCo | Joint/Cartesian Controllers | Robot Manipulation | Kinematics | Inverse Kinematics  [Link](#)

Smart Logistics Mobile-Manipulator Robot: Robot Vision | Mobile Manipulator | Navigation | Manipulation | Python | ROS | Computer Vision  [Link](#)

Haptics-Enabled Smart Goggles for Visually Impaired Users: Computer Vision | YOLOv7 | Embedded Systems | Raspberry Pi | SolidWorks  [Link](#)

Autonomous Maze-Solving Robot Simulation: Robot Vision | Homography | VPR | Pygame | Navigation | Optimal Path Planning  [Link](#)

Skills

Programming Languages: Python, C++, C, MATLAB

Robotics: ROS, ROS 2, MuJoCo, Isaac Sim, Gazebo, MoveIt 2, Robot Kinematics, Robot Dynamics, Motion Planning, Path Planning, Robot Manipulation, SLAM

Machine Learning: PyTorch, Scikit-learn, Neural Networks, Deep Learning, Reinforcement Learning, Imitation Learning, Computer Vision, Data Analysis

Software Packages: SolidWorks, KiCad, Git, Linux, LaTeX, Adobe Creative Suite

Hardware: Raspberry Pi, NVIDIA Jetson Nano, Arduino, ESP32, STM32, M5Stack, Micro:bit, 3D Printing, Laser Cutting, CNC Machining

Languages: English, Nepali